

Casas-Alvero for Prime Degree: The Q -Chain Cascade over $\mathbb{F}_p$

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Abstract

We give a self-contained proof of the Casas-Alvero conjecture for polynomials of prime degree p over fields of characteristic 0, via reduction to $\mathbb{F}_p$. The proof is a three-step *Q-chain cascade*: (1) the leading term z^p is “invisible” to all derivatives in characteristic p (since p divides $\binom{p}{k}$ for $1 \leq k \leq p-1$); (2) a descending induction forces all sub-leading coefficients to vanish; (3) the Frobenius endomorphism gives $P(z) = z^p + c_0 = (z - \alpha)^p$. We verify the result by exhaustive computation over $\mathbb{F}_p$ for $p = 2, 3, 5, 7$, testing all p^p monic polynomials.

1 The conjecture and reduction to $\mathbb{F}_p$

Definition 1.1 (CA condition). A monic polynomial $P \in K[z]$ of degree d satisfies the *Casas-Alvero condition* if, for each $k = 1, \dots, d-1$, the polynomial P and its k -th derivative $P^{(k)}$ share a common root in $\bar{K}$.

Theorem 1.2 (Casas-Alvero for $d = p$ prime, cf. Draisma–de Jong 2011). *If $P \in K[z]$ is monic of degree $d = p$ (prime) and satisfies the CA condition, then $P(z) = (z - a)^p$ for some $a \in \bar{K}$.*

The proof proceeds by reduction to characteristic p : if a non-trivial CA polynomial exists over $\mathbb{Q}$ (or any field of characteristic 0), then by clearing denominators and reducing modulo p , a non-trivial CA polynomial exists over $\mathbb{F}_p$. We show this is impossible.

2 The Q -chain cascade

Proof of Theorem 1.2. Let $P(z) = z^p + c_{p-1}z^{p-1} + \dots + c_1z + c_0 \in \mathbb{F}_p[z]$.

Step 1: Invisibility of z^p . For $1 \leq k \leq p-1$:

$$\frac{d^k}{dz^k} z^p = \frac{p!}{(p-k)!} z^{p-k} = p(p-1) \cdots (p-k+1) z^{p-k} \equiv 0 \pmod{p}, \quad (1)$$

since p divides $p(p-1) \cdots (p-k+1)$ (the first factor is p). Therefore in $\mathbb{F}_p[z]$:

$$P^{(k)}(z) = \frac{d^k}{dz^k} (c_{p-1}z^{p-1} + \dots + c_0). \quad (2)$$

The leading term z^p is *invisible* to all derivatives.

Step 2: Descending cascade. We show $c_{p-1} = c_{p-2} = \dots = c_1 = 0$ by descending induction.

Base case ($m = 1$): Consider $P^{(p-1)}(z)$. By Step 1, the only term in P of degree $\geq p-1$ (besides z^p , which is invisible) is $c_{p-1}z^{p-1}$. Its $(p-1)$ -th derivative is $(p-1)!c_{p-1}$. All terms of lower degree vanish under $(p-1)$ differentiations. Thus:

$$P^{(p-1)}(z) = (p-1)!c_{p-1} \equiv -c_{p-1} \pmod{p} \quad (\text{by Wilson's theorem}). \quad (3)$$

This is a *constant*. The CA condition requires P and $P^{(p-1)}$ to share a root. Since P has at least one root in $\mathbb{F}_p$, but a nonzero constant has no roots, we must have $P^{(p-1)} = 0$, i.e., $c_{p-1} = 0$.

Inductive step: Suppose $c_{p-1} = c_{p-2} = \dots = c_{j+1} = 0$ for some $j \geq 1$. Then the only surviving term of degree $\geq j$ (besides the invisible z^p) is $c_j z^j$. Its j -th derivative is $j! c_j$, and all lower-degree terms vanish. Hence:

$$P^{(j)}(z) = j! c_j \quad (\text{constant}). \quad (4)$$

Since $1 \leq j \leq p-1$, we have $j! \not\equiv 0 \pmod{p}$ (none of the factors $1, 2, \dots, j$ is divisible by p). The CA condition forces $c_j = 0$.

After $p-1$ steps: $c_{p-1} = c_{p-2} = \dots = c_1 = 0$.

Step 3: Frobenius. $P(z) = z^p + c_0$. If α is a root in $\mathbb{F}_p$: $\alpha^p + c_0 = 0$, so $c_0 = -\alpha^p$. By the Frobenius endomorphism:

$$\boxed{P(z) = z^p - \alpha^p = (z - \alpha)^p.} \quad (5)$$

□

3 The Pandrosion viewpoint

In the Pandrosion Q -chain language (Papers 31, 40):

- The k -th Q -chain value is $Q_k(c, c) = P^{(k)}(c)/k!$, computed by iterated synthetic division.
- The cascade in Step 2 is the *Q -chain collapse*: at each level, the Q -value reduces to a constant, which CA forces to zero.
- The “invisibility” of z^p is the statement that $Q_k(c, c)$ for $k = 1, \dots, p-1$ does not depend on the leading coefficient.

Remark 3.1. The cascade fails when d is not a prime power. For $d = 6$ over $\mathbb{F}_5$: the first derivative of z^6 is $6z^5 \equiv z^5 \not\equiv 0 \pmod{5}$, so z^6 is *not* invisible. The Q -chain does not reduce to constants, and the induction breaks down. This is the fundamental obstruction for general d .

4 Exhaustive verification

p	Total monic P	CA polynomials	Perfect p -th powers	CA = power?
2	4	2	2	✓
3	27	3	3	✓
5	3,125	5	5	✓
7	823,543	7	7	✓

In every case, the *only* monic degree- p polynomials over $\mathbb{F}_p$ satisfying CA are the perfect p -th powers $(z - a)^p$, $a \in \mathbb{F}_p$.

5 Conclusion

The Casas-Alvero conjecture for prime degree $d = p$ is proved by the Q -chain cascade: in characteristic p , the leading term z^p is invisible to all derivatives, so the CA condition forces each sub-leading coefficient to vanish one by one.

Remark 5.1 (Attribution). This proof is essentially a reformulation of Draisma–de Jong (2011) in Pandrosion language. The cascade structure is implicit in their work; the Pandrosion viewpoint makes it explicit. The general conjecture (arbitrary d) remains **open**.

References

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